

Targeted Deep Survival Contrasts: Valid Inference for Individualized Treatment Benefit with Neural Networks

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Abstract

Clinical deployment of neural survival models increasingly requires *predictive* claims—how much a treatment would change survival—not just prognostic risk scores. Such claims demand valid inference for counterfactual survival contrasts under confounding and covariate-dependent censoring, targets for which standard deep survival estimators are biased and provide no honest uncertainty. Building on Targeted Deep Architectures (TDA), which embeds TMLE in a network’s weight space, we target the full vector of treatment-specific survival curves $\{S_1(t), S_0(t)\}$ over a time grid, and hence the benefit curve $S_1(t) - S_0(t)$ and the restricted mean survival time (RMST) difference. A single universal targeting update—a ridge projection of the stacked efficient influence functions onto closed-form last-layer gradients—debases every coordinate at once, and a multiplier bootstrap yields simultaneous confidence bands for the whole benefit curve. In 200-replication simulations with confounded treatment, sign-varying effect heterogeneity, and censoring that depends on treatment and covariates, our estimator attains nominal pointwise coverage (95.6%) and near-nominal simultaneous band coverage (94.0%), with benefit-curve bias roughly a third of—and intervals 8% narrower than—a per-timepoint one-step (AIPCW) correction built from the same nuisance fits at comparable coverage; in a companion misspecification study an isotonized one-step closes essentially none of the corresponding gap, so the gain reflects targeting itself rather than monotonicity. The same study delineates the method’s scope: point-estimate double robustness largely survives a badly misspecified outcome network—bias an order of magnitude below the plug-in, though with inflated variance—but the working-model intervals then undercover while the standard one-step stays valid, quantifying the price of superefficiency and when to prefer each estimator.

1 Introduction

Neural networks are now standard tools for individual-level survival prediction from rich clinical inputs, including digital pathology and other high-dimensional modalities. Clinical decisions, however, hinge on *contrasts*: how survival would differ under one treatment versus another. Estimating such contrasts from observational data raises two familiar obstacles—confounding and outcome-dependent censoring mechanisms—and a less appreciated third: a network trained for predictive loss is a biased, inefficient estimator of any low-dimensional causal functional [3, 10], and its naive standard errors are invalid.

Semiparametric theory offers the remedy: debias a flexible initial fit by solving the efficient influence function (EIF) equation, via one-step/AIPCW corrections [3, 6] or targeted maximum likelihood estimation (TMLE) [11]. For survival targets these tools are mature [8, 1, 9], but standard implementations correct *outside* the model: the one-step adds the empirical mean of the EIF to a plug-in, and classical TMLE fluctuates the output distribution after training. Both

become awkward when the target is a whole *vector*—two survival curves on a fine grid—where per-coordinate corrections need not respect monotonicity and can push estimates off the model space. Targeted Deep Architectures (TDA) [7] instead embeds the TMLE fluctuation *inside* the network: a small targeting subset of weights is updated along the projection of the EIF onto per-sample loss gradients, so the fitted network itself is the debiased estimator. TDA was demonstrated for the average treatment effect and the marginal survival curve without treatment; the clinically decisive object—the causal contrast of treatment-specific survival curves under confounding and dependent censoring—was not addressed.

Related work. Neural estimators of treatment-specific survival curves exist—notably SurvITE [5], which targets exactly our estimand with balanced discrete-time hazard networks, and counterfactual time-to-event models [2]—but they provide point estimates without valid confidence statements; that is the gap we fill. Conversely, survival TMLEs with simultaneous inference exist [1, 9] but target the output distribution through analytic fluctuation submodels that become cumbersome for many coordinates and arbitrary architectures; TDA-style weight-space targeting is what removes that obstruction. Causal survival forests [4] give honest inference for conditional effects at fixed horizons rather than the full population benefit curve.

Contributions. We propose *Targeted Deep Survival Contrasts* (TDSC): (1) the target vector $\Psi = (S_1(t_1), \dots, S_1(t_K), S_0(t_1), \dots, S_0(t_K))$ with its stacked EIFs combining treatment and censoring weights (Section 2); (2) closed-form per-sample last-layer gradients for DragonNet-style hazard networks, making the universal TDA update for all $2K$ coordinates one ridge regression per iteration (Section 3); (3) simultaneous sup- t bands for the benefit curve via multiplier bootstrap; (4) simulations with strong confounding, qualitative effect heterogeneity, and treatment- and covariate-dependent censoring, in which TDSC attains nominal coverage with lower bias and narrower intervals than the per-timepoint one-step built from identical nuisance estimates (Section 4).

2 Setup, estimands, and efficient influence functions

Data. We observe n i.i.d. copies of $O = (X, A, \tilde{T}, \Delta)$: covariates $X \in \mathbb{R}^d$, binary treatment $A \in \{0, 1\}$, discrete follow-up time $\tilde{T} \in \{1, \dots, K\}$, event indicator Δ . Time is coarsened into K bins; within a bin, events are recorded before censoring, and subjects surviving bin K are administratively censored. Let $Y(k) = \mathbf{1}\{\tilde{T} \geq k\}$ (at risk) and $dN(k) = \mathbf{1}\{\tilde{T} = k, \Delta = 1\}$ (event), and write $P_n f = n^{-1} \sum_i f(O_i)$. The likelihood factorizes through the propensity $g(X) = P(A = 1 | X)$, the event hazard $h(k | a, x)$, and the censoring hazard $h_c(k | a, x)$, with conditional survival $S(t | a, x) = \prod_{k \leq t} \{1 - h(k | a, x)\}$ and censoring survival S_c defined analogously.

Estimands. Under consistency, no unmeasured confounding, coarsening-at-random censoring (censoring may depend on A and X but not further on the event time), and positivity for treatment and censoring, the treatment-specific curves $S_a(t) = \mathbb{E}_X[S(t | a, X)]$, $a \in \{0, 1\}$, $t \in \{1, \dots, K\}$, are identified. Our target is the $2K$ -vector $\Psi = (S_1(\cdot), S_0(\cdot))$; the *benefit curve* $\beta(t) = S_1(t) - S_0(t)$ and the RMST difference $\Delta_{\text{RMST}} = \sum_{t=1}^K \beta(t)$ (bin width one; the contrast in expected bins survived through horizon K) are linear in Ψ , so their influence functions follow by linearity.

Efficient influence functions. For each coordinate $S_a(t)$ the EIF at P is the discrete-time analogue of Moore and van der Laan [8]:

$$D_{a,t}(O) = - \sum_{k=1}^t \frac{\mathbf{1}\{A=a\}}{g_a(X) S_c(k-1 | a, X)} \frac{S(t | a, X)}{S(k | a, X)} \left(dN(k) - Y(k) h(k | a, X) \right) + S(t | a, X) - S_a(t), \quad (1)$$

with $g_1 = g$, $g_0 = 1 - g$. The first term is a weighted sum of martingale residuals—the within-bin “surprise” of an event relative to the fitted hazard—inversely weighted by the probability of receiving arm a and remaining uncensored through bin $k - 1$; the second is the plug-in centering term. Stacking (1) over a and t gives the $n \times 2K$ influence matrix D whose column means $P_n D_{a,t}$ are exactly the first-order bias terms that targeting drives to zero.

3 Targeted Deep Survival Contrasts

Architecture and initial fit. We use a DragonNet-style network [10] adapted to discrete-time survival: a shared trunk feeding a propensity head for g and two outcome heads producing hazard logits $\{h(k | a, x)\}_{k=1}^K$ for $a \in \{0, 1\}$, trained on the observed-arm discrete-time negative log-likelihood (masked per-bin binary cross-entropy) plus a propensity term, with early stopping. A separate network fits the censoring hazard. Any backbone—including a frozen foundation-model encoder with a trained head—can replace the trunk. Full architecture and hyperparameters are in Appendix A.

Targeting submodel and closed-form gradients. Following TDA, the targeting parameters ϑ are the final linear layers of the two outcome heads, $p = 2K(F+1)$ parameters for head width F . Because those layers are linear in the head features $\phi_a(X)$ and the loss is a per-bin Bernoulli likelihood, the per-sample gradient is available in closed form:

$$\frac{\partial \ell_i}{\partial W_a[k, \cdot]} = \mathbf{1}\{A_i = a\} Y_i(k) (h(k | a, X_i) - dN_i(k)) \phi_a(X_i), \quad (2)$$

and analogously for biases, giving the $n \times p$ score matrix G in one vectorized pass with no per-sample automatic differentiation.

Universal targeting update. Each iteration solves one ridge system for all $2K$ targets, $\alpha = (G^\top G + \lambda I)^{-1} G^\top D$ (with λ scaled to the mean diagonal of $G^\top G$; note $p > n$ here, so the ridge penalty is statistically active), giving per-target directions α_j and projected bias terms $d_j = P_n[G\alpha_j]$. The universal direction is $\alpha^* = \sum_j (d_j / \|d\|_2) \alpha_j$ [7, Sec. 2.3], applied as $\vartheta \leftarrow \vartheta - \gamma \alpha^*$ with a backtracking line search on $\sum_j (P_n D_j)^2$. After each step the hazards and influence matrix are recomputed (g and S_c stay fixed); iteration stops when every $|P_n D_j| \leq \widehat{\text{sd}}(D_j) / (\sqrt{n} \log n)$ or no step improves the criterion (in our simulations the tolerance was never met in full; the line search exited first in every replication—Section 4 reports the diagnostics). Two diagnostics are recorded: the relative projection residual $\|D - G\alpha\| / \|D\|$ (an empirical check of gradient coverage) and the final $\max_j |P_n D_j| / \text{tol}_j$; any coordinate left above tolerance can receive its residual $P_n D_j$ as a one-step top-up, restoring an exactly solved EIF equation. The estimator is the plug-in from the targeted network—by construction a pair of monotone survival curves.

Inference and simultaneous bands. Coordinate standard errors are $\widehat{\text{sd}}(D_j)/\sqrt{n}$ with D evaluated at the final fit; contrasts inherit influence functions by linearity. For the benefit curve we report a sup- t band: with i.i.d. standard normal multipliers ξ_i , the 95% quantile c^* of $\max_t |n^{-1} \sum_i \xi_i D_{\beta,t}(O_i)|/\widehat{\text{se}}_t$ over bootstrap draws gives $\hat{\beta}(t) \pm c^* \widehat{\text{se}}_t$ (c^* floored at 1.96, a conservative choice).

Theoretical guarantees, honestly stated. TDSC is TDA applied to a $2K$ -dimensional pathwise-differentiable target, so Theorems 3.1–3.2 of Li et al. [7]—specializations of adaptive debiased machine learning (ADML) [12]—apply coordinate-wise with a joint CLT. Their conditions require: (T1) regularity of the loss and target; (T2) a first-order expansion, which TDA satisfies by construction *provided sample splitting is used*; (T3) gradient coverage—the span of the final-layer scores approximates the oracle EIF—with controlled empirical-process terms; and (C1)–(C2) negligible oracle bias, under which coverage extends from the working parameter to the true $\Psi(P_0)$. Our implementation does *not* cross-fit: nuisances, targeting directions, and the IF variance all use the full sample, so (T2) as stated is not met and finite-sample validity is supported here by simulation rather than theorem; a cross-fitted variant is mechanical and planned. Two further caveats. First, the EIF (1) is doubly robust—its bias correction survives misspecification of either the hazards or the weights (g, S_c) jointly—but TDSC inherits this only insofar as targeting solves $P_n D_j \approx 0$ (the residual top-up above restores it exactly), and consistent variance estimation still requires the relevant nuisances. Second, when P_0 is well-approximated by the working submodel, ADML yields *superefficiency*: intervals narrower than the nonparametric bound. This is a real benefit in the simulations below, but it is not attainable uniformly over the model—and Section 4 quantifies the price: under gross outcome misspecification the working-model intervals undercover while the one-step’s nonparametric intervals do not.

4 Simulation study

Design. Covariates $X \sim N(0, I_{10})$; treatment $A \mid X \sim \text{Bern}(\text{expit}(0.9X_1 - 0.7X_2 + 0.6X_3))$, truncated to $[0.05, 0.95]$; event hazard $h(k \mid a, x) = \text{expit}(-3.1 + 2k/K + 0.7x_1 + 0.5x_2 + 0.4x_3 - 0.3x_4 + a\tau(x))$ with $\tau(x) = -1 + 0.8x_1 + 0.6x_3$, so the treatment effect changes sign across the population; censoring hazard $\text{expit}(-3.6 + 0.5x_1 + 0.4x_2 + 0.4a + 0.3x_5)$, dependent on treatment and on covariates shared with the outcome (coarsening at random holds; unadjusted estimators are biased). We use $K = 30$, $n = 1000$ ($\approx 61\%$ events, 47% treated in the demo replication). Ground truth is computed by exact integration over 4×10^5 covariate draws; the true Δ_{RMST} is 5.45 bins.

Estimators. (i) **Naive plug-in**: the initial network’s curves; (ii) **unadjusted Kaplan–Meier** by arm; (iii) **one-step (AIPCW)**: per-timepoint $\hat{\Psi}_j^{\text{plug}} + P_n D_j$ at the initial fit, using the same nuisances as TDSC; (iv) **TDSC**, plus its residual top-up variant; (v) in the scenario grid below, an IPTW $\times$ IPCW-weighted KM and an *isotonized* one-step (each arm projected onto monotone curves), which separates how much of TDSC’s advantage is explained by monotonicity alone. Causal survival forests and an output-space survival TMLE [9] are the remaining planned comparators.

Results. Table 2 reports time-averaged operating characteristics over 200 Monte Carlo replications (per-cell coverage MC SE ≈ 1.5 points). Four findings. First, adjustment is essential: unadjusted KM’s absolute bias (0.121) is more than 50-fold TDSC’s (0.0021), and its average Δ_{RMST} misses the truth of 5.45 entirely. Second, the naive plug-in carries the regularization-induced bias that debiasing exists to remove (0.021; RMST bias -0.38). Third, on the benefit curve TDSC and the one-step

both achieve nominal pointwise coverage (95.6% vs 95.1%), but TDSC does so with roughly a third of the bias (0.0021 vs 0.0057) and 8% narrower intervals (0.140 vs 0.151); the MSE point estimates favor TDSC (0.00149 vs 0.00173), though the *paired* replication-level MSE difference (-0.00024 , MC SE 0.00026) is not resolved at 200 replications. The narrower-at-nominal-coverage combination reflects the working-model efficiency of Section 3; the comparison does not isolate weight-space targeting per se from TMLE-versus-one-step, which the isotonized one-step and output-space TMLE comparators in the full study are designed to decompose. Fourth, the two estimators trade off differently on the RMST functional: TDSC is essentially unbiased (0.007 vs -0.161) but has higher variance, giving the one-step the lower RMST MSE (0.53 vs 0.79); both cover at $\approx 95\text{--}97\%$. The simultaneous band covered the whole curve in 94.0% of replications, with $c^* \approx 2.9$. Targeting averaged 9.1 iterations. Figure 1 shows one replication (seed 7; representative—per-replication MAE rankings match the table).

Nuisance misspecification and mechanism. A second study (150 replications per arm; propensity fit by a standalone network throughout so the outcome and weight models can be blinded independently) stresses both arms of double robustness by hiding key covariates from one set of nuisances at a time (Table 1). Three conclusions. *Monotonicity is not the mechanism*: in the well-specified arm the isotonized one-step is indistinguishable from the raw one-step (MSE 0.00154 vs 0.00155), while in this configuration TDSC’s paired MSE advantage is clearly resolved (-0.00052 , MC SE 0.00009)—the gain comes from targeting, not from projecting curves onto the monotone cone. *Point-estimate double robustness holds, with a caveat*: blinding the outcome network to the main confounder and effect modifier explodes the naive plug-in (bias 0.19; RMST bias -5.8) while the EIF-corrected estimators stay an order of magnitude less biased; blinding the weight models instead leaves both corrected estimators protected through the hazard arm while the weighting-only IPTW $\times$ IPCW KM fails (bias 0.076).

Inference is not doubly robust, and TDSC pays first: under outcome misspecification TDSC’s working-model intervals undercover badly (64.6%; top-up 84.3%) while the one-step—whose wider intervals track the nonparametric variance—stays at 95.0%. A *safeguarded* variant—the top-up point estimate paired with the nonparametric (initial-fit) SE—repairs much of the shortfall (coverage 88.8%) at the one-step’s width, but only the one-step itself remains fully nominal, consistent with the targeted fit drifting along a badly misspecified gradient span: even the raw TDSC estimate paired with the nonparametric SE covers only 72%, so the failure lives in point-estimate variability that no IF-based SE captures. Under weight misspecification all corrected estimators undercover mildly ($\approx 89\%$) through the variance estimate. Diagnostics flag the failure in-sample: the mean number of coordinates left above tolerance rises from 37 (well-specified; nominal coverage regardless) to 54 (outcome-misspecified), and the strict $\text{sd}/(\sqrt{n} \log n)$ tolerance was never met in full in any of the 450 replications—the line search exits first—so we recommend reporting unsolved-coordinate counts and applying the residual top-up as defaults.

5 Discussion

TDSC turns a two-headed neural survival network into a valid estimator of population treatment benefit at the cost of a few ridge regressions after training, and the misspecification grid gives the method an honest operating envelope: prefer TDSC when the initial network fits well (narrower intervals at nominal coverage, gains not attributable to monotonicity), and fall back when diagnostics (unsolved-coordinate counts, projection residuals) suggest otherwise—the safeguarded top-up repairs most of the coverage loss, but under gross outcome misspecification only the one-step is fully reliable.

Table 1: Nuisance-misspecification grid, benefit curve ($n = 1000$, $K = 30$, 150 replications per arm; coverage MC SE $\approx 1.8\text{pp}$). *outcome mis.*: outcome hazards blinded to x_1, x_3 ; *weights mis.*: propensity and censoring models blinded to x_1, x_2 . Propensity is fit by a standalone network (not Table 2’s trunk head), so well-specified cells are not directly comparable to Table 2. The isotonized row changes only the point estimate; the safeguard row changes only the interval (its nonparametric SE equals the one-step’s, hence the equal widths).

Scenario	Estimator	Bias	MSE	Coverage	CI width
well-specified	one-step	0.0051	0.00155	94.5%	0.144
	one-step, isotonized	0.0051	0.00154	—	—
	TDSC	0.0036	0.00103	95.7%	0.135
outcome mis.	naive plug-in	0.1922	0.04110	—	—
	one-step	0.0135	0.00345	95.0%	0.197
	TDSC	0.0119	0.01860	64.6%	0.156
	TDSC + top-up	0.0265	0.01325	84.3%	0.156
	+ nonpar. SE (safeguard)	—	—	88.8%	0.197
weights mis.	IPTW $\times$ IPCW KM	0.0760	0.00720	—	—
	one-step	0.0114	0.00111	89.4%	0.111
	TDSC	0.0111	0.00114	89.2%	0.111

The remaining work: (i) cross-fitting, required by the inherited theory, absent from the current implementation, and made more urgent by the variance-estimation failures under misspecification; (ii) an output-space survival TMLE comparator to complete the mechanism decomposition, additional sample sizes, and sensitivity to the ridge penalty, targeting-layer choice, and weight-truncation levels (Appendix A); (iii) *calibration-for-benefit*—targeting arm-specific curves within strata of the network’s own predicted benefit, the estimand used to validate individualized-benefit models against randomized trials—and application on frozen pretrained encoders (e.g. pathology foundation models), which the last-layer construction supports directly.

A Reproducibility

All code, raw Monte Carlo stores, and the exact scripts behind every table are at <https://github.com/blind-contours/tdsc>. Networks: trunk $d \rightarrow 64 \rightarrow 64$ (ELU); propensity head $64 \rightarrow 64 \rightarrow 1$; per-arm outcome heads $64 \rightarrow 64 \rightarrow K$; censoring network $(d+1) \rightarrow 64 \rightarrow 64 \rightarrow (K-1)$. Training: full-batch Adam, lr 10^{-3} , max 500 epochs, early stopping (patience 25) on a 20% validation split, propensity loss weight 1.0. Targeting: ridge scale $\lambda = 0.01 \times \text{mean diag}(G^\top G)$; $\gamma_0 = 1$ with up to 12 halvings; max 50 iterations; tolerance $\widehat{\text{sd}}(D_j)/(\sqrt{n} \log n)$. Truncation: $\hat{g} \in [0.05, 0.95]$, $\hat{S}_c \geq 0.05$, conditional survival floored at 10^{-3} inside (1). Bands: 1000 Gaussian-multiplier draws. No cross-fitting is used (Section 3). Truth: 4×10^5 -draw exact integration.

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Table 2: Benefit curve $S_1(t) - S_0(t)$, time-averaged over 30 points ($n = 1000$, $K = 30$; 200 replications, coverage MC SE $\approx 1.5\text{pp}$). Bias is mean absolute bias. The paired TDSC–one-step MSE difference is -0.00024 (MC SE 0.00026).

	Bias	MSE	Coverage (ptwise)	CI width
Naive neural plug-in	0.0213	0.0026	—	—
Unadjusted KM	0.1212	0.0163	—	—
One-step (AIPCW)	0.0057	0.0017	95.1%	0.151
TDSC (ours)	0.0021	0.0015	95.6%	0.140
Δ_{RMST} : one-step	bias -0.161	MSE 0.53	96.5%	2.87
Δ_{RMST} : TDSC	bias 0.007	MSE 0.79	95.5%	2.69
TDSC simultaneous band	coverage 94.0% (whole curve)		band width 0.207	

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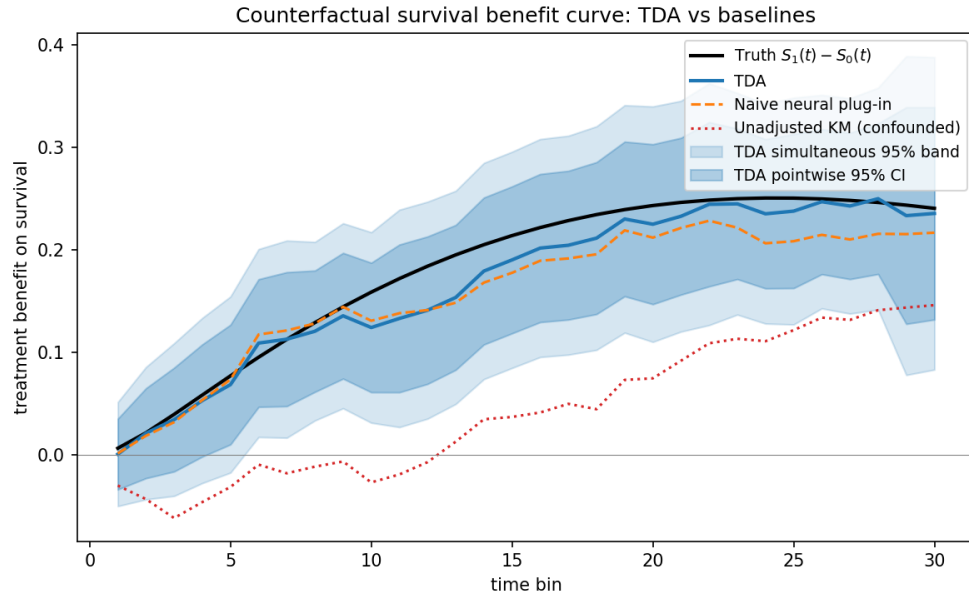


Figure 1: One replication ($n = 1000$, seed 7). The TDSC benefit curve tracks the truth, which lies inside both the pointwise CI and the simultaneous band; the naive plug-in drifts low at later times, and unadjusted KM is severely biased (wrong-signed at early times).